



Figure 1: CHReME layout

posed a serious challenge for the scientists in focusing on their core research area. To make it easier for users who are unfamiliar with the HPC environment but would benefit from using these resources, C-DAC has developed a Web-based tool called CHReME, which is an HPC resource management engine.

CHReME empowers users with an intuitive GUI to exploit HPC resources and provides a layer of abstraction to shield them from the complexity of accessing HPC resources. This enables them to concentrate on their core research/scientific work. CHReME's Web Interface makes clusters of different magnitude easy to manage and monitor, which makes things easier for the HPC systems administrator while enabling researchers and scientists of varied domains to carry out their scientific simulation with minimal efforts.

CHReME enables HPC users to access the cluster resources from any remote location using a Web browser like Mozilla Firefox, Chrome or Internet Explorer, without the need to install any additional software on the client machine.

## CHReME

While sharing the technical details of the portal, Sucheta Pawar from the software development team, HPC Solutions Group, C-DAC Pune, stated that the CHReME portal provides complete access and control of the HPC resources from remote locations by simply logging on to the portal using a standards-compliant Web interface. The portal helps scientists make application-centric job submissions, as well as in monitoring and management. The intuitive cluster administration interface provides statistics regarding cluster utilisation, job accounting, per-user cluster resource use, etc.

To ensure optimal HPC resource use and allocation, CHReME is integrated with workload managers like Torque, Maui, PBSpro, etc, which allows users to submit computationally demanding tasks for transparent and efficient distribution. The user can submit batch jobs, interactive jobs, and parallel jobs to these workload managers through the portal interface. The job status information is intimated to the users via email and portal

alerts, which help them in monitoring the progress of the submitted jobs. CHReME provides a menu-driven interface that is broadly categorised into three key areas, namely, the User Area, the Admin Area and Application Specific Interfaces.

## The User Area

Those wanting to access remote HPC resources do so via a Web-based login interface from any location. The portal uses the SSH network protocol, which provides a secure data communication channel between the two networked systems, i.e., the client machine and the HPC system. CHReME provides an abstract level interface over the SSH network protocol and provides functionalities like a secure shell remote connection, remote shell services or command execution, remote/client side file uploads and downloads, etc.

In short, as the name suggests, the User Area provides various services to end users:

**Job creation and submission:** The creation of application-centric batch script is to define the application execution environment and the entire details of the cluster resources to be used, such as the number of nodes, the processes per node, the job queue, memory requirements, wall time, etc. Submitting jobs to the workload managers ensures proper resource allocation and job execution.

**Job monitoring:** The portal allows users to easily monitor and manage submitted jobs. It provides information about the current status of the submitted job under categories like 'Job Status', 'Job ID', 'Priority', 'Job Name', 'Owner', and 'Queue'. Users can carry out various operations on submitted jobs like *Delete*, *Suspend*, *Resume*, *Hold* and *Release*.

**File uploading and downloads:** The portal interface allows users to remotely upload and download the files. Users can upload the input data, job scripts, etc, and download the output files, post process data, etc.

**Directory structure:** CHReME provides the cluster side directory structure to browse the home directory and its contents like folders and files. Users can create new folders, delete folders/files, rename folders and files, etc.

**Advance reservation:** The portal facilitates advance reservations, which helps users to set aside certain resources to be used over a given timeframe. Users can specify their requirements for computing resources such as selecting the machine for a particular job, the number of processors, and the amount of memory required for running an application, specifying network resources or time slots during which the user expects to utilise the reserved resources.

**Cluster resource information:** The portal provides a list and a graphical view of the cluster resources, allowing users to work with what is available to them at a particular time.